

$$L = (a + b)^+$$

= Regular language

$$27. L = \{w w^R x \mid w \in \{a, b\}^*, x \in \{a, b\}^+\}$$

$$L = (a + b)^+$$

= Regular language

$$28. L = \{w x w^R \mid w \in (a + b)^*, x \in \{a, b\}^+\}$$

$$L = (a + b)^+$$

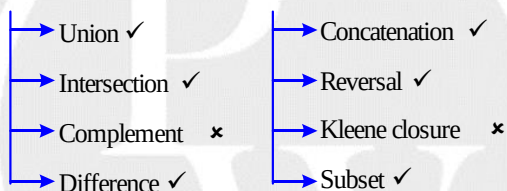
= Regular language

$$29. L = \{x w w^R \mid w \in \{a, b\}^*, x \in \{a, b\}^+\}$$

$$L = (a + b)^+$$

2.17 Closure Properties of Regular Languages

1. Closure properties for finite languages:



2.18 Table for FINITE sets

		Finite sets	Closed/Not Closed
(1)	Union ($\cup$)	$F_1 \cup F_2 \Rightarrow$ Finite	✓
(2)	Intersection ($\cap$)	$F_1 \cap F_2 \Rightarrow$ Finite	✓
(3)	Complement ($\bar{L}$)	$\bar{F} \Rightarrow$ NOT finite	✗
(4)	$L_1 - L_2$	$F_1 - F_2 \Rightarrow$ Finite	✓
(5)	$L_1 \cdot L_2$	$F_1 \cdot F_2 \Rightarrow$ Finite	✓
(6)	L^*, L^+	$F^*, F^+ \Rightarrow$ May or may not be finite	✗
(7)	Subset (L)	Subset (Finite set) $\Rightarrow$ Finite	✓

2.18.1 Table for Infinite sets

		Infinite sets
(1)	Union ($\cup$)	Infinite $\cup$ Infinite $\Rightarrow$ Infinite
(2)	Intersection ($\cap$)	Infinite $\cap$ Infinite $\Rightarrow$ Need not be infinite
(3)	Complement ($\bar{L}$)	Complement of Infinite $\Rightarrow$ May or may not be infinite
(4)	$L_1 - L_2$	Need not be infinite
(5)	$L_1 \cdot L_2$	Infinite
(6)	L^*, L^+	Infinite
(7)	Subset (L)	Need not be infinite

2.19 Closure Properties of Regulars

$L_i \rightarrow$ Regular

1.	$\cup$	2.	$\cap$
3.	$\bar{L}$	4.	$L_1 - L_2$
5.	$L_1 \cdot L_2$	6.	L^{Rev}
7.	L^+	8.	L^+
9.	Subset (L) is not closed for regular languages	10.	Prefix (L)
11.	Suffix (L)	12.	Substring (L)
13.	Substitution (L)	14.	Homomorphism (L)
15.	ϵ -free homomorphism (L)	16.	$h^{-1}(L)$ (Inverse homomorphism (L))
17.	L_1/L_2 (Quotient)	18.	Symmetric difference
19.	Half (L)	20.	Second half (L)
21.	One-third (L) [$1/3(L)$]	22.	Middle $1/3(L)$
23.	Last $1/3(L)$	24.	New $(L_1, L_2) = \text{Suffix}(L_1 \cup L_2^+)$
25.	Finite Union	26.	Finite Intersection
27.	Finite Difference	28.	Finite Concatenation

29.	Finite Subset	30.	Finite Substitution
31.	Infinite Union	32.	Infinite $\cap$
33.	Infinite Difference	34.	Infinite Concatenation
35.	Infinite Subset	36.	Infinite Substitution

- Out of the given 36 closure properties, how many are closed for regular languages?

Subset operation, and 6 infinite operations are not closed, remaining all are closed for regular languages.

Out of 36 operations total 7 operations are not closed.

2.19.1 Operations over regular languages

Examples:

$$(1) \quad \left. \begin{array}{l} L_1 = a^* \\ L_2 = a^+ \end{array} \right\} \Rightarrow L_1 \cdot L_2 = a^*$$

$$(2) \quad \left. \begin{array}{l} L_1 = \phi \\ L_2 = \text{Any} \end{array} \right\} \Rightarrow L_1 + L_2 = L_2$$

$$(3) \quad \left. \begin{array}{l} L_1 = a^* \\ L_2 = b^* \end{array} \right\} \Rightarrow L_1 + L_2 = a^* + b^*$$

$$(4) \quad \left. \begin{array}{l} L_1 = \Sigma^* \\ L_2 = \text{Any language over same } \Sigma \end{array} \right\} \quad \begin{array}{l} L_1 + L_2 = \Sigma^* \\ = L_1 \end{array}$$

$$(5) \quad \left. \begin{array}{l} L_1 = a\Sigma^* \\ L_2 = b\Sigma^* \end{array} \right\} \quad \Sigma = \{a, b\} \quad \begin{array}{l} L_1 + L_2 = a\Sigma^* + b\Sigma^* \\ = (a + b)\Sigma^* \\ = \Sigma^+ \end{array}$$

Note :

$$1. \quad \left. \begin{array}{l} L_1 = \phi \\ L_2 = \text{Any} \end{array} \right\} \Rightarrow L_1 \cdot L_2 = \phi$$

$$2. \quad \left. \begin{array}{l} L_1 = \Sigma^* \\ L_2 = \text{Any} \end{array} \right\} \Rightarrow L_1 + L_2 = L_2$$

2.19.2 Properties of regular languages

I. If both L_1 and L_2 are regular sets then $L_1 \cap L_2$ is Regular.

II. IF $L_1 \cap L_2$ is regular set then

- L_1 “need not be regular”

- L_2 “need not be regular”.

III. If L is regular then $\bar{L}$ is regular.

IV. If $\bar{L}$ is regular then L is regular.

V. L is regular iff $\bar{L}$ is regular

VI. L is not regular iff $\bar{L}$ is not regular.

Example:

$a^n b^n$ is not regular and $\overline{a^n b^n}$ is not regular.

2.19.3 Arden's Lemma and Kleene Method

Arden's Lemma:

If $R = Q + RP$ and P does not contain ϵ then $R = QP^*$

$$R = Q + RP \quad \dots(1)$$

$$= Q + (Q + RP)P = Q + QP + RP^2 \quad \dots(2)$$

Substitute R one more time in equation (2)

$$R = Q + QP + QP^2 + RP^3$$

If we do repetitive substitution infinite time, we will get $R = QP^*$

Kleene Method:

Kleene Method

→ Dynamic programming $O(n^3)$
→ R_{ij}^k where i is the initial state, j is the final state
k represents number of states

$$R_{ij}^k = R_{ij}^{k-1} + R_{ik}^{k-1} (R_{kk}^{k-1})^* R_{kj}^{k-1}$$

Regular Grammar

→ It represents a regular set
→ It can be either left linear grammar (LLG) or right linear grammar (RLG)

LLG	RLG
Each production in LLG follows $V \rightarrow VT^*$ OR $V \rightarrow T^*$	Each production in RLG follows $V \rightarrow T^*V$ OR $V \rightarrow T^*$

2.20 Identify the Language Generated by Regular Grammar

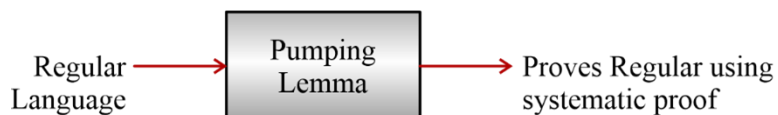
	Regular Grammar	Regular Language
1.	$S \rightarrow \epsilon$	$L = \{\epsilon\}$
2.	$S \rightarrow \epsilon a$	$L = \{\epsilon, a\}$
3.	$S \rightarrow aa abc d$	$L = \{aa, abc, d\}$

4.	$S \rightarrow Aa$ $A \rightarrow b$	By default S is a start symbol here. $L = \{ba\}$
5.	$S \rightarrow Aa$	Useless production. It has no meaning. $L = \{\} = \phi$
6.	$S \rightarrow \boxed{Ab} \mid b$ $\downarrow$ Useless	$L = \{b\}$
7.	$\uparrow S \rightarrow Aa \mid Ba$ $A \rightarrow a$ $B \rightarrow b$ Look from bottom to top	$L = \{aa, bb\}$
8.	$\uparrow S \rightarrow Aa$ $A \rightarrow \epsilon \mid b$	$L(A) = \{\epsilon, b\}$ $L = L(S) = L(A) \cdot a$ $= \{a, ba\}$

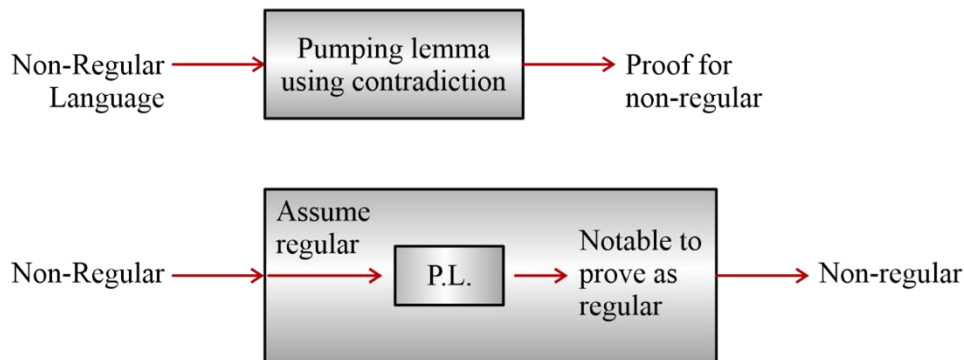
Regular Language	Regular Grammar
a^*	(I) $S \rightarrow Sa \mid \epsilon$ OR (II) $S \rightarrow aS \mid \epsilon$
a^+	(I) $S \rightarrow Sa \mid a$ OR (II) $S \rightarrow aS \mid a$
$(a + b)^*$	(I) $S \rightarrow Sa \mid Sb \mid \epsilon$ OR (II) $S \rightarrow aS \mid bS \mid \epsilon$
$(a + b)^+$	(I) $S \rightarrow Sa \mid Sb \mid a \mid b$ OR (II) $S \rightarrow aS \mid bS \mid a \mid b$
$(ab)^*$	(I) $S \rightarrow Sab \mid \epsilon$ OR (II) $S \rightarrow abS \mid \epsilon$

2.21 Pumping-Lemma (PL)

2.21.1 Pumping Lemma for Regular Languages

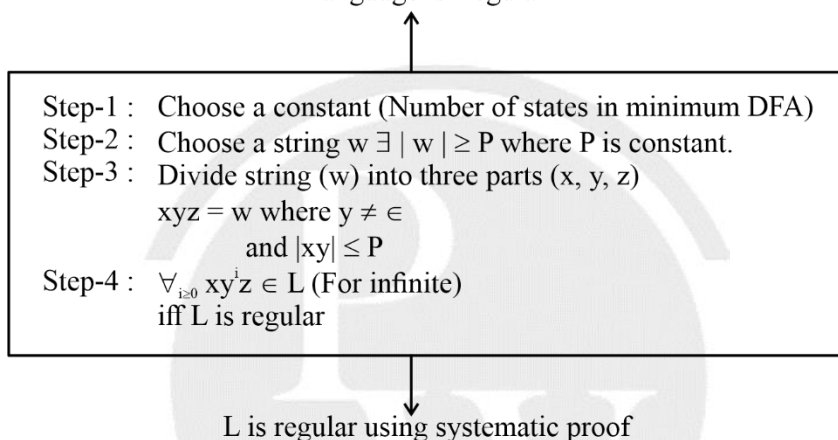


2.21.2 Pumping Lemma for Non-Regular Languages using contradiction



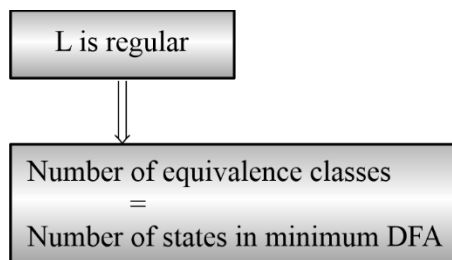
2.21.3 Proof for Regular Languages

- If L is Regular language, then how pumping lemma proves it as regular?
Language is Regular



2.22 Equivalence Classes

- L is regular iff L has finite number of equivalence classes.
- L is not regular iff L has infinite equivalence classes.



□□□